

STUDENT PERFORMANCE **PREDICTION**

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Abstract

Academic achievement is a cornerstone of a student's educational journey, influencing future career opportunities and personal growth. Predicting student performance enables educational institutions to identify and support students who may be at risk of underperforming. This report presents a comprehensive machine learning pipeline — Logistic Regression, Decision Tree, Random Forest, k-Nearest Neighbors, and XGBoost — applied to the UCI Student Performance dataset. Beyond the baseline comparison, this iteration of the project adds engineered features, stratified cross-validation, hyperparameter tuning, SHAP-based explainability, and an ablation study that quantifies how much predictive power comes specifically from prior academic grades versus demographic and behavioral factors. A tuned **XGBoost** model achieved the best cross-validated performance (macro F1 ≈ 0.90), and the full pipeline is packaged as a reusable, deployable project with an interactive Streamlit demo.

1 Introduction

Student performance is a primary indicator of educational effectiveness and is critical for forecasting future academic and professional success. Educational institutions increasingly rely on data-driven approaches to identify students requiring intervention. Predictive analytics powered by machine learning offers a robust framework to classify students by risk level and design targeted support strategies.

Academic institutions often face challenges in predicting student success due to the multifaceted nature of educational performance, influenced by cognitive, behavioral, social, and psychological factors. Traditional analytical methods often fall short in capturing these complexities. Machine learning, by contrast, offers a more sophisticated approach by detecting hidden patterns and making data-driven predictions. This project explores the feasibility of predicting student performance through machine learning methods, and — going beyond the original coursework scope — investigates *which* predictions are actually useful in practice by separating the contribution of prior grades from other, earlier-available signals.

Scope of the Report

This report investigates the application of supervised machine learning algorithms to predict student performance using a dataset comprising demographic, academic, and behavioral attributes. The goal is to provide actionable insights for educators, administrators, and policy-makers, enabling data-informed strategies to support student achievement, and to demonstrate an end-to-end, production-style ML workflow (versioned pipeline, tuned models, explainability, and a deployable interface).

Problem Statement

Academic performance significantly affects students' life trajectories. However, predicting performance is challenging due to the multitude of influencing factors such as demographics, socio-economic background, behavioral habits, and psychological parameters. The core problem addressed in this project is: *“Can we predict students' academic performance accurately based on available demographic, academic, and behavioral attributes — and how much of that accuracy depends on already knowing their prior grades?”*

Objectives

- Predict students' academic performance category (Low / Medium / High) from available attributes.
- Identify key determinants impacting academic success, with quantified feature importance.
- Evaluate and compare multiple ML models using robust cross-validation, not a single split.
- Tune the strongest candidates and quantify the gain from hyperparameter optimization.
- Separate the predictive contribution of prior grades from behavioral/demographic factors (ablation study).
- Provide actionable, explainable insights for educators via SHAP analysis.
- Package the work as a reusable, deployable, GitHub-ready project.

2 Literature Review

Recent years have seen a surge in research focused on predicting student performance using machine learning and data mining techniques. Studies have explored a variety of algorithms, including Decision Trees, Random Forests, Support Vector Machines, Naive Bayes, and Neural Networks, each with varying degrees of success.

Key findings from the literature:

- **Feature Selection:** Prior academic performance, attendance, participation, and parental background are consistently identified as strong predictors.
- **Model Performance:** Ensemble methods like Random Forest and boosting algorithms often outperform single classifiers due to their ability to handle complex, non-linear relationships.
- **Explainability:** Recent work emphasizes model transparency, using techniques such as SHAP values to interpret feature importance and build stakeholder trust. This report directly applies SHAP to the final model (Section 6).
- **Personalized Interventions:** Predictive models are increasingly used to design targeted interventions, such as personalized study recommendations and early warning systems.

Despite these advances, challenges remain in data quality, ethical considerations, and generalizability across diverse educational contexts — points revisited in Section 10.

3 Methodology

3.1 Data Collection

The project uses the UCI Student Performance Dataset (Mathematics course), comprising 395 students' demographic information, academic records, and lifestyle habits, collected from two secondary schools in Portugal (Cortez & Silva, 2008).

Attributes included:

- **Demographic:** age, sex, address, family size, parental education, job types.
- **Academic:** grades (G1, G2, G3), study time, failures, school support.
- **Behavioral:** absences, extracurricular activities, free time, alcohol consumption, health status.

3.2 Feature Engineering

Beyond the raw UCI columns, three engineered features were added to give models (and this analysis) more structured signal:

- `avg_alcohol` — mean of workday (`Dalc`) and weekend (`Walc`) alcohol consumption.
- `parent_edu_avg` — mean of mother's and father's education level.
- `avg_prior_grade` — mean of `G1` and `G2`, capturing overall prior trend.

3.3 Data Preprocessing

- **Missing values:** none present in the original dataset.
- **Encoding:** implemented via a single `scikit-learn` `ColumnTransformer` inside a `Pipeline` — one-hot encoding for binary and nominal categorical columns, standard scaling for numeric columns. This replaces ad-hoc `LabelEncoder` usage from the original version, which risked implying false ordinal relationships between nominal categories (e.g. parental job types) and could leak scaling statistics across CV folds.
- **Target construction:** `G3` (0–20) is bucketed into *Low* (0–9), *Medium* (10–14), and *High* (15–20).
- **Splitting:** stratified 80/20 train-test split, with all model comparison done via stratified 5-fold cross-validation on the training set.

3.4 Exploratory Data Analysis

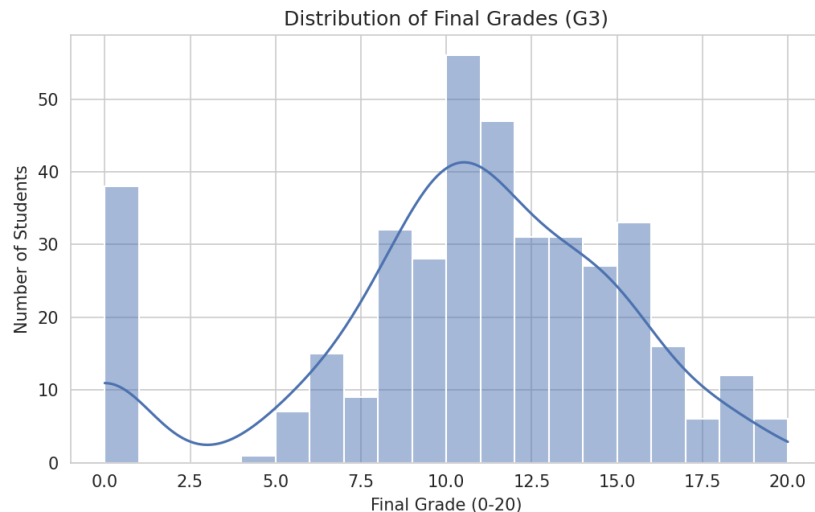


Figure 1: Distribution of final grades (G3) across all 395 students.

Key EDA findings: strong positive correlation between `G1`, `G2`, and `G3`; a negative relationship between absences and final grade; higher weekend alcohol consumption associated with lower grades; and higher study time associated with better outcomes.

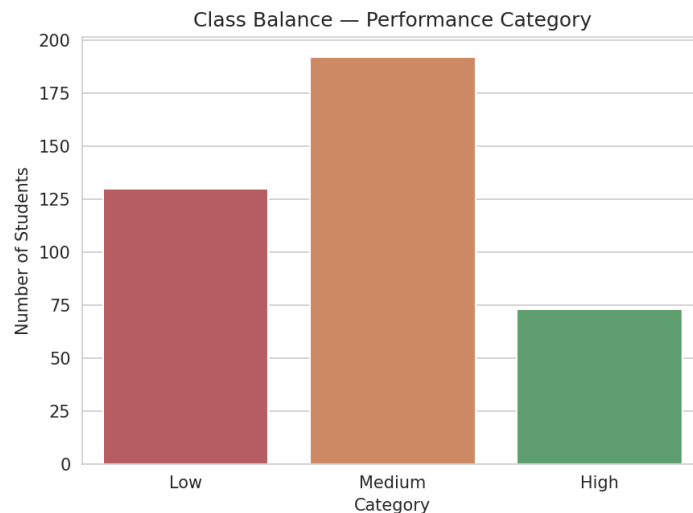


Figure 2: Class balance after bucketing G3 into Low / Medium / High. The dataset is moderately imbalanced (Low: 130, Medium: 192, High: 73), which is why macro-averaged F1-score — not raw accuracy — is used as the primary tuning and comparison metric.

4 Model Building

4.1 Pipeline Design

All five candidate models are wrapped in an identical `Pipeline(preprocessor, classifier)` so that encoding and scaling are fit *only* on training folds during cross-validation — eliminating the data leakage risk present when preprocessing is done once on the full dataset before splitting.

4.2 Baseline Comparison (5-fold Stratified Cross-Validation)

Model	CV Accuracy	CV Macro F1
Logistic Regression	0.836 ± 0.019	0.834 ± 0.020
Decision Tree	0.842 ± 0.022	0.844 ± 0.025
Random Forest	0.867 ± 0.045	0.869 ± 0.042
k-Nearest Neighbors	0.706 ± 0.050	0.696 ± 0.056
XGBoost	0.886 ± 0.043	0.892 ± 0.038

Table 1: Baseline 5-fold stratified CV results (untuned models, training set only).

XGBoost and Random Forest are the clear front-runners, confirming the literature’s consistent finding that ensemble/boosting methods outperform single classifiers on this type of structured, moderately-sized tabular data.

4.3 Hyperparameter Tuning

The two strongest candidates were tuned using `GridSearchCV` (5-fold CV, scored on macro F1):

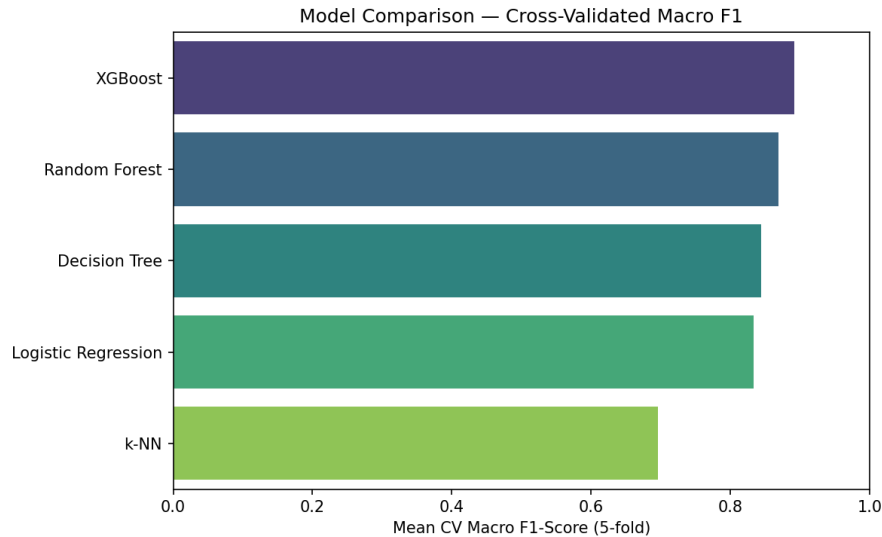


Figure 3: Cross-validated macro F1-score comparison across all baseline models.

- **Random Forest** — grid over `n_estimators` $\in \{100, 200, 400\}$, `max_depth` $\in \{\text{None}, 6, 10, 16\}$, `min_samples_split` $\in \{2, 4, 8\}$, `min_samples_leaf` $\in \{1, 2, 4\}$. **Best:** `n_estimators=100`, `max_depth=None`, `min_samples_split=4`, `min_samples_leaf=1` $\rightarrow$ **CV Macro F1** = 0.891.
- **XGBoost** — grid over `n_estimators` $\in \{100, 200, 300\}$, `max_depth` $\in \{3, 4, 6\}$, `learning_rate` $\in \{0.05, 0.1, 0.2\}$. **Best:** `n_estimators=300`, `max_depth=4`, `learning_rate=0.1` $\rightarrow$ **CV Macro F1** = 0.899.

Tuned **XGBoost** was selected as the final model.

5 Model Evaluation

5.1 Held-out Test Set Performance

Class	Precision	Recall	F1-score
High	0.94	1.00	0.97
Low	0.81	0.85	0.83
Medium	0.89	0.84	0.86
Accuracy	0.873		
Macro avg	0.88	0.90	0.89
Weighted avg	0.87	0.87	0.87

Table 2: Classification report — tuned XGBoost on the held-out 20% test set (79 students).

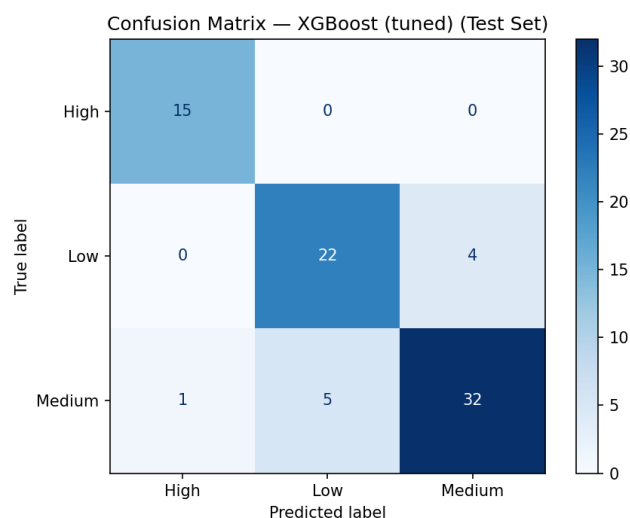


Figure 4: Confusion matrix — tuned XGBoost, held-out test set. All 15 “High” students were correctly identified; most confusion occurs between the adjacent Low/Medium classes, which is expected given they are artificial buckets of a continuous grade.

5.2 Ablation Study: Predicting Without Prior Grades

G1 and G2 dominate feature importance in essentially every student-performance study, including this one — but that also makes the prediction less actionable: if a student’s period grades are already known, a model isn’t required to notice they’re struggling. The more useful question is whether performance can be flagged *before* G1/G2 exist, using only demographic and behavioral data.

Feature set	Test Accuracy	Test Macro F1
With G1/G2 (full model)	0.899	0.915
Without G1/G2 (ablation)	0.557	0.473

Table 3: Ablation study — Random Forest trained with vs. without prior-grade features.

As expected, performance drops substantially when G1/G2 are removed: test accuracy falls from 89.9% to 55.7%, a drop of approximately 34 percentage points. However, the model still performs above the majority-class baseline ($\approx 49\%$) using only demographic, behavioral, and lifestyle features. This supports a two-stage early-warning design: a lighter model for *early-semester* risk flagging before grades exist, and the full model for *mid-year* intervention targeting once G1/G2 are available.

6 Feature Importance and Explainability

To move beyond raw feature-importance scores (which can be misleading for correlated or one-hot-encoded features), SHAP (SHapley Additive exPlanations) values were computed for the final XGBoost model.

Reading the plot: G2 and the engineered `avg_prior_grade` dominate, confirming the correlation analysis. Beyond grades, weekend alcohol consumption (`WalC`), absences, and age

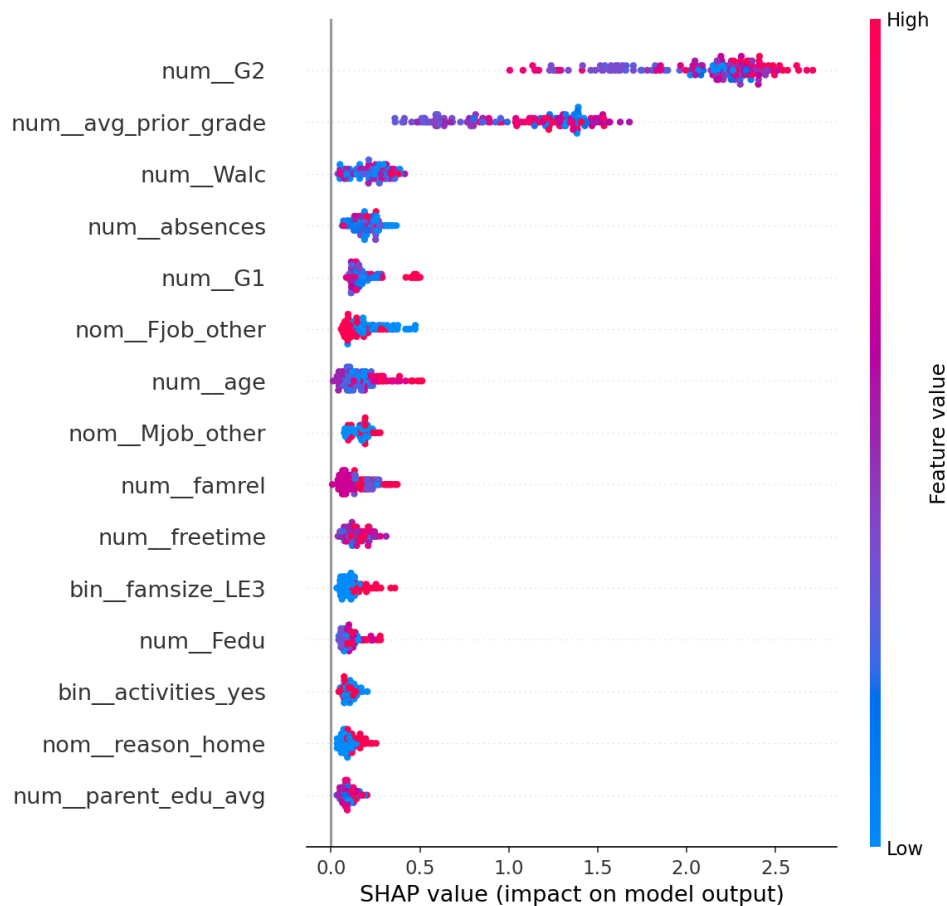


Figure 5: SHAP summary plot — top 15 features by mean absolute SHAP value (averaged across the three output classes). Point color indicates the feature’s value (red = high, blue = low) and horizontal position shows its impact on the model’s output.

carry meaningful secondary signal — consistent with the discussion in Section 8 and the wider literature on behavioral predictors of academic outcomes.

7 Results Summary

- **Best model:** Tuned XGBoost — CV Macro F1 = 0.899, Test Accuracy = 87.3%, Test Macro F1 = 0.888.
- **Key predictors:** G2, avg_prior_grade, G1, absences, weekend alcohol use (Walc), age.
- **Ablation finding:** removing prior grades reduces Random Forest test accuracy from 89.9% to 55.7%, a drop of approximately 34 percentage points, demonstrating the strong predictive contribution of prior academic performance.
- **Hyperparameter tuning gain:** CV Macro F1 improved from 0.892 (default XGBoost) to 0.899 (tuned) — a modest but consistent gain, showing the defaults were already reasonably close to optimal for this dataset size.

8 Discussion

- **Academic history dominates:** Previous grades (G1, G2) are the strongest predictors, aligning with educational theory that past performance is a reliable indicator of future

success.

- **Behavioral factors matter, but secondarily:** absences and alcohol use are significant contributors, but the ablation study shows they cannot substitute for the signal carried by prior grades.
- **Ensemble/boosting methods win:** XGBoost and Random Forest both outperform Logistic Regression, Decision Tree, and k-NN — confirming their suitability for this kind of structured, mixed categorical/numeric tabular data.
- **Practical implication:** a two-tier early-warning system is plausible — a lightweight model using only demographic/behavioral data early in the term, upgraded to the full model once period grades become available.

9 Deployment

To make this project usable beyond a notebook, the trained pipeline is wrapped in:

- **src/predict.py** — a reusable inference function and CLI for single-student predictions.
- **app.py** — an interactive Streamlit web app where a user adjusts sliders/dropdowns for a hypothetical student and receives a live Low/Medium/High prediction with class probabilities.

The full source, trained model, notebook, and this report are published at: <https://github.com/tanvirkaur30/student-performance-prediction>

10 Limitations

- The dataset is limited to 395 students from two schools in Portugal, which may limit generalizability to other educational systems.
- Psychological and socio-emotional factors were not captured in the original data and could further improve prediction accuracy.
- The Low/Medium/High bucketing is a modeling choice; a regression formulation of G3 was not explored here but is a natural extension.
- Ethical considerations — data privacy, and the risk of a predictive label influencing how a student is treated by staff — must be addressed before any real-world deployment.

11 Conclusion

Predicting student performance using machine learning is highly feasible: a tuned XGBoost model reached 87.3% test accuracy and 0.888 macro F1 on a held-out set. Prior academic grades remain the dominant predictor, but the ablation study shows meaningful signal persists in purely demographic and behavioral data, supporting early-intervention use cases. SHAP analysis confirms the model's reasoning aligns with domain intuition (grades, attendance, and lifestyle factors), which is important for building institutional trust in such a system.

Future work could explore regression on the raw G3 score, incorporate mental-health or online-learning-behavior indicators, and validate the approach across additional schools or countries before any operational deployment.

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